

Jatin Madan

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University of California, Irvine — Master of Computer Science — GPA: 4.0 [Irvine, CA, USA] Sept 2023 - Dec 2024
Vellore Institute of Technology — B.Tech Computer Science and Engineering [Vellore, India] July 2016 - June 2020

SKILLS

Languages	Python, C#, PowerShell, Bash, SQL, Java, JavaScript, Swift
Security Engineering	Workload Identity (OIDC, OAuth 2.0, SAML), Secrets Management, RBAC, Kubernetes Security (Namespaces, Network Policies, Admission Controllers, Pod Security), CI/CD Supply Chain Security (GHAS, Signed Attestations, Dependency Verification), SAST, IaC Security Scanning, Zero Trust, Threat Modeling, CSPM
Cloud & Infra	Azure (AKS, Key Vault, Entra ID, Function Apps, VNet, Log Analytics, VMSS), AWS (S3, Lambda), Multi-Cloud IAM, VPC Architecture, Network Segmentation, Terraform, Bicep, Docker, Kubernetes, GitHub Actions, Azure DevOps
Tools	Microsoft Sentinel, Defender for Cloud, Purview, Databricks, Power BI, Postman, PySpark, Pandas, NLTK, Django

EXPERIENCE

Deloitte — Engineering Manager — Cloud Infra and Security Team — Seattle, WA Jan 2025 - Present

- Built and shipped an **MCP server on AKS** implementing **OIDC-based workload identity, credential rotation, and secrets management via Azure Key Vault** — enabling secure service-to-service **Microsoft Graph API** operations end-to-end, from identifying the auth gap to monitoring in production
- Engineered **Kubernetes cluster security controls** including **RBAC policies, namespace isolation, pod security standards, network policies, and admission controllers** across multi-tenant AKS environments supporting **150K+ user workloads**
- Hardened **CI/CD pipelines** against supply chain attacks by integrating **GitHub Advanced Security (GHAS), signed attestations, dependency verification, automated IaC scanning, and policy enforcement** across **2,000+ application** deployment pipelines in GitHub Actions and Azure DevOps
- Developed **secure development libraries and migration accelerators** in **Python and PowerShell** — standardizing **credential issuance, RBAC configuration, and identity replication** across Entra ID tenants so engineering teams could ship identity-safe code by default, reducing manual effort by **70%**
- Implemented **AI security and governance frameworks** including **red-teaming, model guardrails, and policy-driven monitoring** via **Purview and Defender for Cloud**, motivated by a deep belief in the role security engineering plays in building trustworthy and safe AI systems
- Contributed to **continuous CSPM** using **Terraform IaC scanning, misconfiguration detection, and automated remediation** — identified gaps through code review and threat modeling, shipped fixes, and maintained **100% policy compliance** across multi-cloud production environments

ZS Associates — Associate Consultant Intern — Cloud and Big Data Team — Philadelphia, PA Jun 2024 - Aug 2024

- Implemented **Unity Catalog** in Databricks with **fine-grained RBAC, encryption, and audit logging** across **Azure Data Lake Storage** — took ownership end-to-end from identifying access control gaps to shipping and monitoring the governance solution across 3,000+ tables
- Built and published a **hardened Docker image** via **Artifactory** with security-vetted, pre-installed packages, eliminating runtime installation attack surface across all Databricks clusters and reducing provisioning time by **14%**
- Identified and remediated **cloud security misconfigurations** through hands-on infrastructure code review, producing documented findings and collaborating with the platform team to ship fixes within the sprint cycle

University of California, Irvine — Data & Tech Fellow — Prof. Tom Andriola — Irvine, CA Jan 2024 - Dec 2024

- Engineered a **hybrid workload identity system** using **Entra ID Connect, OIDC federation, Application Proxy, and HRD policies** — implemented **credential issuance and rotation** patterns across a multi-provider environment, debugging authentication flows end-to-end
- Built an **Apple Vision OS remote development platform** in **Swift** (SSH tunneling, VS Code Server, sandboxed execution) — designed to support **rapid-iteration research workflows** with secure credential management and network isolation as first-class requirements
- Performed **threat modeling and hands-on debugging** of authentication and session flows, identified token replay and session hijacking vectors, and shipped fixes validated through code review — taking full ownership from discovery through monitoring

- Built an **Automated Multi-Cloud Security Assessment Tool** in **Python** scanning for **IAM misconfigurations, network segmentation gaps, and RBAC violations** across multi-tenant Azure environments — reducing manual effort by **78%** and improving misconfiguration detection by **35%**
- Implemented **Zero Trust network architecture** in Azure — engineering **VPC segmentation, east-west traffic controls, IAM policy enforcement, encryption in transit and at rest, and VNet isolation** across multi-cloud production environments
- Developed an **Azure DevOps IaC scanning extension in Python** that **pre-scanned Terraform templates** for supply chain policy violations and misconfigurations before deployment — integrated end-to-end from detection to automated remediation in CI/CD
- Overhauled **Kubernetes and PaaS IAM** by automating **RBAC role mapping, secrets rotation, and identity lifecycle management** via **Python and PowerShell** tooling; designed **secure service-to-service authentication patterns** using **OAuth 2.0 and SAML** across distributed Azure workloads
- Shipped a **Secure SSO system** and **Video Conferencing Platform** applying **OAuth 2.0 authentication, API security controls, and serialization hardening** — collaborating across infrastructure and application teams to make secure patterns the path of least resistance

CERTIFICATIONS		LINK TO ALL 11 CERTIFICATIONS		
• Microsoft Certified: Azure Security Engineer Associate 2021	AZ-500	H736-2347	Mar 28,	
• Microsoft Certified: Azure Solutions Architect Expert 2021	AZ-304	H693-3978	Feb 21,	
• Microsoft Certified: DevOps Engineer Expert 2021	AZ-400	H813-4148	May 16,	
• Microsoft Certified: Azure Developer Associate 2021	AZ-204	H717-3092	Mar 13,	
• Microsoft Certified: Azure Network Engineer Associate 2021	AZ-700	I483-3372	Nov 12,	